# Cube-Sum Permutations: An Explicit Finite-Certificate Proof of Eventual Existence and Multiplicity

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**Abstract**

Write $P(n)$ for the number of permutations $p$ of $[n] = \{1, \dots, n\}$ for which $i + p(i)$ is an integer cube for every $i$. We prove $P(n) > 0$ for every $n \geq 173$ and determine exactly which smaller indices admit such permutations, resolving the eventual-membership question for OEIS A006063. We also show $P(173) = P(174) = 1$ and $P(n) \geq 3$ for every $n \geq 175$, yielding the complete membership of the related multiplicity sequence A096680. The infinite existence argument is symbolic: an interval-overlap reflection lemma propagates a verified finite interval to all larger indices. For the finite base, a universal block-rearrangement identity converts 56 specified root-word/order pairs into 784 positive integer blocks by exact determinants, supplemented by six algebraic block families and a forced-neighbor argument at $n = 173$. The small negative and multiplicity classifications use independently checkable finite Hall and alternating-cycle certificates. We further prove that every odd $n \geq 69603$ has a cube-sum involution fixing only 4, and every even $n \geq 228028$ has a fixed-point-free cube-sum involution.

## 1 Problem and results

Let $\mathcal{C} = \{k^3 : k \in \mathbb{N},\ k \geq 2\}$ and let $P(n)$ count bijections $p : [n] \to [n]$ satisfying

$$i + p(i) \in \mathcal{C} \qquad (1 \leq i \leq n). \tag{1}$$

The card-arranging problem was discussed by Gardner [1, 2] and included in Sloane and Plouffe's sequence encyclopedia [3]. Gardner's later account credits Angeloo Papaikonomou with early work on the cube-sum variant and reports a uniqueness observation; OEIS records Ray Chandler's counterexamples and Robert Israel's 2018 eventual-existence conjecture [2, 4, 5]. The counting function $P(n)$ is OEIS A096904 [6].

Write $\mathcal{G} = \{n \geq 1 : P(n) > 0\}$. The increasing enumeration of $\mathcal{G}$ is OEIS A006063 [4]; the indices supporting at least two distinct bijections form OEIS A096680 [5]. We count permutations of $[n]$, not just involutions: symmetry of the allowed sums does not force a perfect matching to be symmetric.

**Theorem 1.1** (Exact existence classification)**.** *For every $n \geq 173$, $P(n) > 0$. The admissible values below 173 are exactly*

$$\begin{aligned}\mathcal{G} \cap [1, 172] = \{&7, 19, 26, 37, 44, 56, 63, 66, 68, 80, 82, 85, 87, 98, 100,\\ &103, 105, 110, 112, 115, 116, 117, 118, 119, 120, 121, 122, 123,\\ &124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 135, 147, 149,\\ &150, 151, 152, 155, 156, 159, 171\}.\end{aligned} \tag{2}$$

*In the usual OEIS indexing,* A006063$(j) = j + 124$ *for $j \geq 49$.*

**Theorem 1.2** (Multiplicity classification)**.** *Set $E = \{112, 115, 116, 117, 119, 124, 125, 126, 127, 128, 129, 130, 133\}$. Then*

$$\{n \geq 1 : P(n) \geq 2\} = E \cup \{175, 176, 177, \dots\}.$$

*Moreover, $P(173) = P(174) = 1$, $P(175) = 3$, and $P(n) \geq 3$ for every $n \geq 175$. Consequently* A096680$(j) = j + 161$ *for $j \geq 14$.*

The proof is computer-assisted through exact finite certificates; the infinite propagation is symbolic. Sections 2–4 give the existence proof; Sections 5–6 establish the two exact classifications. Section 7 proves the eventual involution theorem, and Section 8 specifies the supplementary data and verification procedures. Appendices A and B give the algebraic families and critical peeling rounds.

## 2 Reflection and an infinite interval avalanche

We temporarily admit the empty permutation on $[0] = \varnothing$ as a seed, but $P(0)$ is not an OEIS term.

**Lemma 2.1** (Cubic suffix reflection)**.** *Let $0 \le m < n$ and suppose $P(m) > 0$, with $P(0) = 1$ by convention. If $c = m + n + 1$ is a cube, then $P(n) > 0$. The extension sends $i \in [m+1, n]$ to $c - i$ and agrees with the original permutation on $[m]$.*

*Proof.* Since $c - (m+1) = n$ and $c - n = m + 1$, the new suffix is mapped bijectively onto itself. Its index–image sums all equal $c$, while the first $m$ assignments remain valid. □

**Lemma 2.2** (Interval-overlap avalanche)**.** *Suppose every integer of $[a, b]$ is admissible, where $1 \le a \le b$. If the integer cube $c$ satisfies*

$$a + b + 2 \le c \le 2b + 2, \tag{3}$$

*then every integer in $[a, c - a - 1]$ is admissible.*

*Proof.* For a new integer $n \in [b+1, c-a-1]$ set $m = c - n - 1$. Inequality (3) gives $m \ge a$ and $m \le b$; also $m < n$ since $c \le 2b + 2 \le 2n$. Thus Lemma 2.1 applies to each such $n$. The old interval remains covered. □

**Theorem 2.3** (The finite interval implies the entire tail)**.** *If $[173, 364] \subseteq \mathcal{G}$, then $[173, \infty) \subseteq \mathcal{G}$ without further finite certificates.*

*Proof.* For $(a, b, c) = (173, 364, 729)$ we have $539 \le 729 \le 730$, and Lemma 2.2 gives $[173, 555] \subseteq \mathcal{G}$. More generally, suppose $k \ge 9$ and $[173, k^3 - 174] \subseteq \mathcal{G}$. The next cube $c = (k+1)^3$ obeys

$$k^3 + 1 \le (k+1)^3 \le 2k^3 - 346.$$

For the second inequality the difference is $f(k) = k^3 - 3k^2 - 3k - 347$. Now $f(9) = 112$ and $f(k+1) - f(k) = 3k^2 - 3k - 5 > 0$ for $k \ge 9$. Lemma 2.2 extends the interval to $[173, (k+1)^3 - 174]$. The upper endpoints tend to infinity. □

**Remark 2.4** (A Hall obstruction)**.** *Coverage by domains of cube reflections alone is insufficient: at $n = 65$ the reflection domains for 64 and 125 cover $[1, 65]$, but $S = \{3, 40, 59, 64\}$ has only $\{5, 24, 61\}$ as possible images, violating Hall's inequality.*

## 3 Block rearrangements and integer determinant certificates

For positive integers $\ell_1, \dots, \ell_t$ summing to $N$, partition $[N]$ into consecutive input blocks $I_1, \dots, I_t$ of those lengths. Let $\pi$ be a permutation of the labels $[t]$, prescribing the order of blocks of the same lengths on the output side. Write

$$A_j = \sum_{i<j} \ell_i, \quad B_j = \sum_{\mathrm{pos}_\pi(i) < \mathrm{pos}_\pi(j)} \ell_i, \quad C_j = 1 + A_j + B_j + \ell_j,$$

where $\mathrm{pos}_\pi(j)$ is the location of label $j$ in the output order.

**Lemma 3.1** (Reflected-block rearrangement)**.** *If every $C_j$ is a cube, the formula $p(i) = C_j - i$ for $i \in I_j$ defines a cube-sum permutation of $[N]$. Zero-length blocks may be omitted simultaneously from both input and output orders; they impose no condition on their nominal $C_j$.*

*Proof.* The input block $I_j = [A_j + 1, A_j + \ell_j]$ is sent in reverse order to $[B_j + 1, B_j + \ell_j]$, its assigned output block. Since the output blocks partition $[N]$, the piecewise map is a bijection. Its sums are exactly $C_j$ on each nonempty block. □

A common parametrization eliminates the need to store endpoints. Let $k_1, \dots, k_t \ge 2$ be integer cube roots, and write $d_j = k_j^3 - 1$. Define the integer matrix

$$M(\pi)_{ji} = \mathbf{1}_{i<j} + \mathbf{1}_{\mathrm{pos}_\pi(i) < \mathrm{pos}_\pi(j)} + \mathbf{1}_{i=j}. \tag{4}$$

Then the block-sum equations $C_j = k_j^3$ are precisely $M(\pi)\ell = d$.

**Proposition 3.2** (Exact integer certificate)**.** *For a specified root word $k$ and order $\pi$, suppose $\det M(\pi) = 1$. Set*

$$\ell_j = \det\big(M(\pi) \text{ with column } j \text{ replaced by } d\big). \tag{5}$$

*If all the $\ell_j$ are positive, this defines a cube-sum permutation on $N = \sum_j \ell_j$.*

*Proof.* Cramer's rule over the integers gives $M(\pi)\ell = d$, so the prescribed reflected block sums are cubes. Lemma 3.1 proves the conclusion. □

## 4 The finite starting interval

The supplementary root-word/order table contains 56 rows of the form

$$N \mid k_1, \ldots, k_t \mid \pi_1, \ldots, \pi_t.$$

Together they specify 784 nonempty reflected input blocks. For each row, exact integer arithmetic verifies the determinant, positive block lengths, total $N$, cube-sum identities and output coverage.

**Proposition 4.1** (Algebraic seed data)**.** *The 56 certified root-word/order pairs all satisfy* $\det M = 1$, $\ell_j \geq 1$, *and* $\sum_j \ell_j = N$. *Six additional explicit block-template families yield 25 further seed lengths:*

$$66, 85, 100, 103, 119, 122, 131, 132, 150, 151, 155, 156,$$
$$174, 175, 182, 228, 229, 231, 238, 240, 241, 254, 345, 351, 357.$$

*The 56 determinant shapes cover 10 seed lengths below 173 and 46 in* $[173, 364]$*; the 25 templates cover 12 and 13, respectively. Each template instance is an application of Lemma 3.1, including the rule for omitted zero-length blocks.*

*Proof.* For each of the 56 rows, the finite data satisfy the hypotheses of Proposition 3.2. Appendix A gives the six block-template identities and their parameter choices, which produce the 25 remaining lengths by Lemma 3.1. □

The remaining seed $n = 173$ follows from forced-neighbor elimination. On a finite vertex set $V$, put a loop at $i$ if $2i$ is a cube, and an undirected edge between distinct $i, j$ if $i + j$ is a cube.

**Lemma 4.2** (Forced-neighbor elimination)**.** *If a vertex* $i$ *has exactly one neighbor* $j$ *in this graph, every cube-sum permutation forces* $p(i) = j$ *and* $p(j) = i$. *Removing the one loop or the two endpoints gives a bijection between permutations before and after the deletion.*

*Proof.* Row $i$ has only column $j$ available. By symmetry, column $i$ has only row $j$ available. Both assignments are forced; restriction and unique extension are inverse operations. □

**Proposition 4.3** (Critical base and exact first counts)**.** *There is exactly one cube-sum permutation at* $n = 173$ *and at* $n = 174$, *while* $P(175) = 3$.

*Proof.* Appendix B lists the complete forced-neighbor rounds for $n = 173$, 174 and 175, starting with three uniform bands. Each listed endpoint has exactly one neighbor in the remaining vertex set. At $n = 173$ and 174 the residual graph is empty, yielding uniqueness by Lemma 4.2. At $n = 175$ the residual graph has exactly three permutations, as shown in Appendix B.3. The forced deletions preserve the count, so $P(175) = 3$. □

**Proposition 4.4** (Exact finite reflection closure)**.** *Starting with the empty permutation, the 56 certified determinant seeds, the 25 algebraic template seeds, and the forced-neighbor seed at 173, repeated application of Lemma 2.1 in increasing order constructs 48 lengths below 173 and* every *one of the 192 lengths* $173 \leq n \leq 364$.

*Proof.* There are 22 direct seeds below 173 and 60 direct seeds in $[173, 364]$, totaling $56 + 25 + 1 = 82$. The following deterministic recursion is the complete remaining finite check. For each $n = 1, \ldots, 364$ in order, declare $n$ admissible if it is a seed or there exists a previously marked $m$ with $0 \leq m < n$ and $m + n + 1 \in \{8, 27, 64, 125, 216, 343, 512\}$. The exact audit returns 26 additional admissible lengths below 173 and 132 in $[173, 364]$. In particular, each $n \in [256, 338]$ follows from $m = 511 - n \in [173, 255]$ by the cube 512. All seed and derived lengths have explicit bijections by the preceding lemmas. The finite check verifies both the size of the closure and inclusion of the entire target interval. □

**Corollary 4.5.** *For every* $n \geq 173$ *there is a cube-sum permutation of* $[n]$.

*Proof.* Combine Proposition 4.4 with Theorem 2.3. □

## 5 Exact classification below the threshold

For $S \subseteq [n]$ let

$$\mathcal{N}_n(S) = \{j \in [n] : \exists i \in S,\ i + j \in \mathcal{C}\}.$$

If $|\mathcal{N}_n(S)| < |S|$, Hall's theorem [7] forbids a cube-sum permutation. The supplementary small-index table partitions $[1, 172]$ into exactly 48 records with explicit valid permutations and 124 records with explicit Hall-deficient sets. The verifier checks bijectivity and each permitted sum for positive records and the complete neighbor set for every negative record. The Hall deficiencies are strict (1 to 6 for the recorded sets). Thus all 172 sizes are classified.

*Proof of Theorem 1.1.* The 48 positive and 124 negative certificates establish exactly (2). Corollary 4.5 supplies all subsequent integers. Therefore the 49th sequence entry is 173, and thereafter the $j$th is $j + 124$. □

## 6 Multiplicity through alternating cycles

For a valid permutation $p$ on $[n]$ form a directed graph $D_p$ on $[n]$ by drawing $i \to j$ exactly when $i \neq j$ and $i + p(j) \in \mathcal{C}$.

**Lemma 6.1** (Alternating-cycle criterion)**.** *A distinct valid permutation $q \neq p$ exists if and only if $D_p$ contains a directed cycle. The cycle $i_1 \to \cdots \to i_s \to i_1$ gives a new valid permutation by replacing $p(i_r)$ by $p(i_{r+1})$ for all $r$ cyclically.*

*Proof.* The replacement preserves the multiset of images and all new sums are allowed. Conversely, for any $q \neq p$, the permutation $p^{-1} \circ q$ has a nontrivial cycle, and every edge of that cycle is present in $D_p$. □

The supplementary cycle tables specify distinct alternative permutations relative to the reference permutations identified in Section 8. The 48 admissible indices below 173 comprise 13 with certified cycles (the set $E$ of Theorem 1.2) and 35 whose directed alternation graphs are acyclic. The former admit multiple permutations; the latter are unique by Lemma 6.1. The forced-neighbor argument independently certifies uniqueness at 173 and 174.

For each of the 190 sizes $175 \leq n \leq 364$, the two recorded cycles produce *three* distinct permutations in total. The same holds for $n = 554, 555$ by two separate cycles after reflecting the unique permutations at $174, 173$ under cube 729. These assertions are finite and explicitly checked, including distinctness of the two alternatives.

**Lemma 6.2** (A cube-gap estimate)**.** *For every $n \geq 556$ some cube $c$ belongs to $[n + 176, 2n]$.*

*Proof.* Let $k$ be the least integer with $k^3 \geq n + 176$; then $k \geq 10$ and $n \geq (k-1)^3 - 175$. Consequently

$$2n - k^3 \geq k^3 - 6k^2 + 6k - 352 \geq 0.$$

The final polynomial equals 108 at $k = 10$ and increases thereafter, since its forward difference is $3k^2 - 9k + 1 > 0$ for $k \geq 10$. □

*Proof of Theorem 1.2.* The 13 positive small multiple indices, the 35 unique small indices and uniqueness at 173 and 174 give the exact initial classification. We now propagate the three-permutation lower bound. For $365 \leq n \leq 553$, take $c = 729$ and $m = 728 - n \in [175, 363]$, so the three distinct permutations at $m$ extend distinctly to $n$. The exceptional $n = 554, 555$ have independent finite certificates as described above. For $n \geq 556$, Lemma 6.2 supplies $c \in [n + 176, 2n]$, so $m = c - n - 1 \in [175, n - 1]$; induction and Lemma 2.1 finish the proof. The set $E$ has 13 elements, so the 14th multiple-size index is 175 and all subsequent indices are consecutive, giving the stated OEIS formula. □

## 7 Eventual involutions with the fewest fixed points

An *involution* is a permutation $p$ satisfying $p^2 = \mathrm{id}$. Although the preceding existence theorem does not require involutivity, strict cube reflections also give an independent large-index result.

**Theorem 7.1** (Eventual parity-minimal involutions)**.** *For every odd integer $n \geq 69603$ there is a cube-sum involution on $[n]$ whose unique fixed point is 4. For every even integer $n \geq 228028$ there is a cube-sum involution on $[n]$ with no fixed points. In particular, the minimum number of fixed points permitted by parity is attained on both of these infinite tails.*

*Proof.* An odd-cube reflection extension from length $m$ to length $n$ as in Lemma 2.1 preserves parity because $m + n + 1$ is odd. It extends an involution to an involution: the new suffix is invariant under $i \mapsto m + n + 1 - i$, and there are no new fixed points since the reflection sum is odd. Start with the two explicit involutions

$$p_7(i) = 8 - i \quad (1 \leq i \leq 7), \qquad p_{26}(i) = 27 - i \quad (1 \leq i \leq 26).$$

The first fixes precisely 4; the second is fixed-point-free.

The finite portion of the proof is an integer interval calculation. Fix either seed $a \in \{7, 26\}$, its parity $\varepsilon = a \bmod 2$, and a cap $H$. Write $n = 2t + \varepsilon$. Extension using an odd cube $c = k^3$ sends the source coordinate $t$ to

$$t' = d_k - t, \qquad d_k = \frac{k^3 - 1 - 2\varepsilon}{2}.$$

For a known interval $t \in [A, B]$, strict extension $t' > t$ and the bound $t' \leq T = (H - \varepsilon)/2$ give the exact source interval

$$L = \max(A, d_k - T), \qquad U = \min\bigl(B, \lfloor (d_k - 1)/2 \rfloor\bigr).$$

Whenever $L \leq U$, the new certified integer interval is $[d_k - U, d_k - L]$. Start with the singleton coordinate of $a$, apply this rule simultaneously, and merge adjacent or overlapping integer intervals. The resulting finite arithmetic certificates are shown below; the rightmost interval is displayed in the original length coordinate.

| seed | $H$ | rounds | intervals | start | final rightmost interval |
|---|---|---|---|---|---|
| 7 | 100001 | 13 | 2961 | 69603 | $[69603, 100001]$ |
| 26 | 303412 | 12 | 7047 | 228028 | $[228028, 303412]$ |

Every certified length has the fixed-point structure of its seed. Thus the theorem holds throughout the indicated initial parity windows.

It remains to propagate past the caps. For $(M, H) = (69603, 100001)$ in the odd case or $(228028, 303412)$ in the even case, take any $n > H$ of the required parity and let $k$ be the least *odd* positive integer with $k^3 \geq n + M + 1$. Put $m = k^3 - n - 1$. Then $m \geq M$ and $m \equiv n \pmod 2$. We show $m < n$. Minimality of $k$ implies $n \geq (k-2)^3 - M$, hence

$$2n - k^3 \geq 2(k-2)^3 - k^3 - 2M =: F_M(k).$$

The choices of $H$ ensure that $k \geq 57$ in the odd case and $k \geq 83$ in the even case. At their respective initial roots,

$$F_{69603}(57) = 8351 > 0, \qquad F_{228028}(83) = 35039 > 0.$$

Moreover $F_M(k+2) - F_M(k) = 6k^2 - 36k + 8 > 0$ for $k \geq 57$. Consequently $k^3 \leq 2n$, so $M \leq m < n$. Strong induction within each parity and an odd-cube suffix reflection establish the claims for every larger $n$. Parity forces every odd-length involution to have at least one fixed point; an even-length one may have none. □

**Remark 7.2.** *The bounds* 69603 *and* 228028 *are sufficient and are not asserted to be the least indices where arbitrary cube-sum involutions exist. The allowed cube sums grow with* $n$.

## 8 Reproducibility

The supplementary data and exact-integer verifiers are available in the companion repository: https://github.com/dongxuelian2/oeis-papers/tree/main/papers/A006063. The source, data and code used for a verification run are identified by SHA256 hashes in the generated report; the supplementary manifest is `audit/SUPPLEMENT.sha256`.

1. **Existence.** The 56 determinant rows are in `certificates/finite_56_cubic_order_shapes.txt`. The checker `verify_56_plus_forcing.py` reconstructs their block lengths, generates the 25 template instances and verifies the increasing reflection closure through 364. The complete forced-neighbor rounds in Appendix B are checked by `verify_forced_peeling.py`.

2. **Small classification.** `certificates/exact_classification_below_173.json` contains the 48 positive permutations and 124 Hall-deficient sets for $1 \leq n \leq 172$.

3. **Multiplicity.** The cycles in `certificates/second_solutions.json` and `certificates/third_solutions.json` refer to the small positive permutations just described and, for $175 \leq n \leq 364$, the permutations in `certificates/base_permutations.json` at zero-based index $n - 173$. For $n = 554, 555$, the reference permutations are the cube-729 extensions of that archive's permutations at 174 and 173, respectively. The checker `verify_extended.py` validates the reference permutations, Hall deficits, acyclicity tests and the distinctness of all alternatives.

4. **Involutions.** The checker `verify_eventual_involutions.py` reconstructs the two finite parity windows in Theorem 7.1 from the elementary seeds.

With Python 3 and assertions enabled, the aggregate command, run from `papers/A006063`, executes these checks and writes their results:

```
python3 reproduce.py --report verification.json
```

The computations use exact integers. The infinite conclusions follow from the reflection and cube-gap arguments in the preceding sections.

## A Six algebraic block families

For each family, $\ell$ lists the input lengths, $\pi$ lists the output block labels, and $C$ lists the nominal sums $C_j = 1 + A_j + B_j + \ell_j$ of Lemma 3.1. Zero-length blocks are deleted from both orders; only the sums on nonempty blocks need to be cubes. The following formulas and parameter choices give the 25 seed lengths in Proposition 4.1.

### A.1 Four, five and six blocks

**Family I.** For integer cubes $U < V$, put

$$a = 2U - V - 1, \qquad b = 2V - 3U + 1.$$

If $a, b > 0$, take

$$\ell = (a, b, a, b), \qquad \pi = (3, 4, 1, 2), \qquad C = (U, V, U, V).$$

The total length is $N = 2(V - U)$. The three choices $(U, V) = (64, 125), (125, 216), (216, 343)$ give, respectively, $(a, b) = (2, 59), (33, 58), (88, 39)$ and $N = 122, 182, 254$.

**Family II.** For integer cubes $X < Y < Z$, put

$$a = 2Y - Z - 1, \qquad b = X + 2Z - 4Y + 1, \qquad c = Y - X.$$

If $a, b, c > 0$, take

$$\ell = (a, b, c, a, c), \qquad \pi = (4, 2, 5, 1, 3), \qquad C = (Y, X, Z, Y, Z).$$

Here $N = 2Y - X - 1$. With $(Y, Z) = (64, 125)$, the choices $X = 27, 8$ give $(a, b, c) = (2, 22, 37), (2, 3, 56)$ and $N = 100, 119$.

**Family III.** For integer cubes $X, W, Y, Z$, put

$$a = 2Y - Z - 1, \qquad b = X - 2a - 1,$$
$$c = W + 2a - 2X + 1, \qquad d = Y + X - W - 2a - 1.$$

If $a, b, c, d > 0$, take

$$\ell = (a, b, c, d, a, d), \qquad \pi = (5, 2, 3, 6, 1, 4), \qquad C = (Y, X, W, Z, Y, Z).$$

Then $N = 2Z - 2Y + X - W$. With $(Y, Z) = (64, 125)$, the choices $(X, W) = (8, 64), (27, 64), (8, 27)$ give $(a, b, c, d) = (2, 3, 53, 3), (2, 22, 15, 22), (2, 3, 16, 40)$ and $N = 66, 85, 103$, respectively.

*Proof.* For each displayed output order, substitution in $C_j = 1 + \sum_{i<j} \ell_i + \sum_{\mathrm{pos}_\pi(i)<\mathrm{pos}_\pi(j)} \ell_i + \ell_j$ gives the indicated vector $C$ and total $N$. All its entries are cubes and the listed lengths are positive, so Lemma 3.1 applies. □

### A.2 Twelve blocks with three optional blocks

**Family IV.** Choose integers $X, T, Q$ and put

$$u = X - 7, \qquad v = T - 2X + 7, \qquad w = X + 27 - T,$$
$$z = Q + 2T - 2X - 121, \qquad h = X + 152 - T - Q.$$

Assume $u, v, z \geq 0$ and $w, h > 0$. Set

$$\ell = (3, u, v, w, 3, w, z, h, 3, 27, 3, h),$$
$$\pi = (11, 2, 3, 6, 9, 4, 7, 12, 5, 10, 1, 8),$$
$$C = (125, X, T, 64, 125, 64, Q, 216, 125, 216, 125, 216).$$

The total length is $N = 276 + X - T - Q$. Require $X$ to be a cube when $u > 0$, $T$ to be a cube when $v > 0$, and $Q$ to be a cube when $z > 0$. Choose $X \in \{7, 8\}$, then $T \in \{2X - 7, 27\}$, and then $Q \in \{121 - 2T + 2X, 125\}$. The eight instances are

| $X$ | $T$ | $Q$ | $u$ | $v$ | $z$ | $N$ |
|---|---|---|---|---|---|---|
| 7 | 27 | 125 | 0 | 20 | 44 | 131 |
| 8 | 27 | 125 | 1 | 18 | 42 | 132 |
| 8 | 9 | 125 | 1 | 0 | 6 | 150 |
| 7 | 7 | 125 | 0 | 0 | 4 | 151 |
| 7 | 7 | 121 | 0 | 0 | 0 | 155 |
| 8 | 9 | 119 | 1 | 0 | 0 | 156 |
| 8 | 27 | 83 | 1 | 18 | 0 | 174 |
| 7 | 27 | 81 | 0 | 20 | 0 | 175 |

*Proof.* The prefix-sum identity gives the displayed $C$ and $N$. For these choices $w \in \{7, 8, 27, 26\}$ and $h > 0$ by substitution. If $X = 7$ then $u = 0$; if $T = 2X - 7$ then $v = 0$; and if $Q = 121 - 2T + 2X$ then $z = 0$. Otherwise the respective sums are the cubes 8, 27 and 125. Every other nominal sum is a cube. Deleting the zero blocks therefore gives the required permutations. □

### A.3 Fifteen blocks with two binary parameters

**Family V.** Let $\alpha, \beta \in \{0, 1\}$ with $\alpha \leq \beta$, put $t_0 = 7 + 4\beta + 2\alpha$, and take $t \in \{0, t_0\}$. Define

$$(\ell_1, \ldots, \ell_5) = (2\beta,\ 7 - 2\alpha - 4\beta,\ \alpha,\ \alpha,\ 13 + 4\beta),$$
$$(\ell_6, \ldots, \ell_{10}) = (\alpha,\ 23 - 4\beta - \alpha,\ 13 + 4\beta,\ 48 - 45\beta,\ 2\beta),$$
$$(\ell_{11}, \ldots, \ell_{15}) = (\beta(39 - \alpha),\ t,\ 30 + \alpha - t,\ 13 + 4\beta,\ 91 + \alpha - t),$$
$$\pi = (10, 4, 6, 2, 8, 3, 11, 1, 9, 14, 7, 12, 15, 5, 13).$$

Using $\alpha\beta = \alpha$, the nominal sums are

$$(C_1, \dots, C_5) = (21 + 43\beta,\ 8,\ 28 - \alpha,\ 8,\ 216),$$
$$(C_6, \dots, C_{10}) = (21 + 2\alpha + 4\beta,\ 125,\ 64,\ 125,\ 105 - 41\beta),$$
$$(C_{11}, \dots, C_{15}) = (125,\ 209 - 2\alpha - 4\beta + t,\ 343,\ 216,\ 343).$$

The total length is $N = 238 + \alpha + 2\beta - t$. Thus

| $\alpha$ | $\beta$ | $t$ | $N$ |
|---|---|---|---|
| 0 | 0 | 0 | 238 |
| 0 | 0 | 7 | 231 |
| 0 | 1 | 0 | 240 |
| 0 | 1 | 11 | 229 |
| 1 | 1 | 0 | 241 |
| 1 | 1 | 13 | 228 |

*Proof.* The lengths are nonnegative for the six listed choices. Substitution in the prefix-sum identity gives the displayed $C$. Blocks 1, 10 and 11 are nonempty only when $\beta = 1$, with sums $64, 64, 125$. Blocks 3, 4 and 6 are nonempty only when $\alpha = \beta = 1$, with sums $27, 8, 27$. The sum of block 12 is needed only when $t = t_0$, and then equals 216. The remaining blocks have sums $8, 216, 125, 64, 125, 343, 216, 343$. All sums on nonempty blocks are therefore cubes, so Lemma 3.1 applies. □

### A.4 Ten blocks with nested binary parameters

**Family VI.** Let $\alpha, \beta \in \{0, 1\}$ with $\beta \le \alpha$. Set

$$(\ell_1, \dots, \ell_5) = (7,\ 12\alpha,\ 159 - 24\alpha - 99\beta,\ 14\beta,\ 28\beta),$$
$$(\ell_6, \dots, \ell_{10}) = (63\beta,\ 10 + 24\alpha - 6\beta,\ 8\beta,\ 159 - 24\alpha - 107\beta,\ 10 + 24\alpha + 93\beta),$$
$$\pi = (1, 2, 8, 5, 6, 9, 4, 10, 3, 7).$$

The nominal sums and total length are

$$(C_1, \dots, C_5) = (8,\ 15 + 12\alpha,\ 343,\ 333 - 24\alpha - 93\beta,\ 174 - 49\beta),$$
$$(C_6, \dots, C_{10}) = (174 + 42\beta,\ 512,\ 184 + 24\alpha + 8\beta,\ 343,\ 512),$$
$$N = 345 + 12\alpha - 6\beta.$$

The choices $(\alpha, \beta) = (0, 0), (1, 0), (1, 1)$ give $N = 345, 357, 351$, with five, six and ten nonempty blocks, respectively.

*Proof.* Direct substitution gives the displayed nominal sums, and all lengths are nonnegative. Blocks 1, 3, 7, 9 and 10 are always present, with sums $8, 343, 512, 343, 512$. Block 2 is present only at $\alpha = 1$, when its sum is 27. Blocks 4, 5, 6 and 8 are present only at $\beta = 1$, hence $\alpha = 1$, with sums $216, 125, 216, 216$. Lemma 3.1 gives all three permutations. □

## B Complete forced-neighbor rounds at the critical sizes

In a remaining vertex set $V$, the notation $S : c$ means that every $i \in S$ has the unique neighbor $c - i$ in $V$. Remove $i$ and $c - i$ for all such $i$ simultaneously, counting a loop only once. A semicolon separates groups in the same round. All neighbor tests below use the six cubes $\{8, 27, 64, 125, 216, 343\}$; no larger cube can occur when $n \le 181$.

### B.1 The first three rounds

For $173 \le n \le 181$, use

| round | $S : c$ | image interval |
|---|---|---|
| 1 | $[125, 342 - n] : 216$ | $[n - 126, 91]$ |
| 2 | $[34, 215 - n] : 64$ | $[n - 151, 30]$ |
| 3 | $[95, 276 - n] : 216$ | $[n - 60, 121]$ |

*Proof.* For round 1, $i \ge 125$ excludes cubes at most 125, while $343 - i \ge n + 1$; thus only $216 - i$ is available. For round 2, the alternative $125 - i$ is in the deleted image interval $[n - 126, 91]$, while $216 - i > n$; cubes at most 27 are too small. For round 3, the alternative $125 - i$ lies in the deleted interval $[n - 151, 30]$ and $343 - i > n$; cubes at most 64 are too small. In each round the displayed source and image intervals are disjoint and still present. Their sizes are $218 - n$, $182 - n$ and $182 - n$, respectively, so these rounds remove $1164 - 6n$ vertices. □

### B.2 The remaining rounds for $n = 173$

After the first three rounds, continue with the following table. Each row is one round, not a simultaneous grouping of successive rows.

| round | $S : c$ |
|---|---|
| 4 | $\{4, 5\} : 8$; $[8, 12] : 27$ |
| 5 | $\{45, 46, 106, 107, 108, 122\} : 216$ |
| 6 | $\{31\} : 64$; $\{172, 173\} : 216$ |
| 7 | $\{92\} : 216$ |
| 8 | $\{1\} : 8$ |
| 9 | $\{20\} : 125$ |
| 10 | $\{111\} : 125$ |
| 11 | $\{13\} : 125$ |
| 12 | $\{104\} : 125$ |
| 13 | $\{6\} : 8$ |
| 14 | $\{123\} : 216$ |
| 15 | $\{32\} : 64$ |

At every round, subtracting each listed endpoint from the six possible cubes shows that its only remaining neighbor is the displayed partner. These rounds exhaust [173]. The loops occur at 4, 108 and 32; all other removals are reciprocal pairs. Lemma 4.2 therefore gives $P(173) = 1$.

### B.3 The remaining rounds for $n = 174$ and $n = 175$

Round 4 is $\{4\} : 8$; $[8, 11] : 27$ for $n = 174$, and $[8, 10] : 27$ for $n = 175$. For both sizes the subsequent rounds are

| round | $S : c$ |
|---|---|
| 5 | $\{45, 46, 47, 106, 107, 108\} : 216$ |
| 6 | $\{15\} : 27$; $\{172, 173, 174\} : 216$ |
| 7 | $\{113\} : 216$ |
| 8 | $\{22\} : 27$ |
| 9 | $\{3\} : 125$ |
| 10 | $\{94\} : 125$ |
| 11 | $\{33\} : 125$ |
| 12 | $\{124\} : 125$ |
| 13 | $\{7\} : 27$ |
| 14 | $\{105\} : 216$ |
| 15 | $\{14\} : 27$ |
| 16 | $\{112\} : 216$ |
| 17 | $\{21\} : 27$ |
| 18 | $\{2\} : 125$ |
| 19 | $\{93\} : 125$ |

Again each listed endpoint has exactly one remaining neighbor, verified by subtraction from the six cubes. For $n = 174$ the residual set is empty, with loops at 4 and 108, so $P(174) = 1$. For $n = 175$, precisely the ten vertices

$$\{4, 11, 16, 23, 41, 48, 102, 114, 168, 175\}$$

remain. Testing their pairwise sums gives the loop at 4, the edge $4 \sim 23$, and the chordless cycle

$$23 \sim 41 \sim 175 \sim 168 \sim 48 \sim 16 \sim 11 \sim 114 \sim 102 \sim 23.$$

Its consecutive edge sums are $64, 216, 343, 216, 64, 27, 125, 216, 125$. If 4 is fixed, the odd cycle has the two orientations as its only permutation covers: reciprocal pairs cannot cover an odd number of vertices. If 4 is paired with 23, the remaining path has a unique cover by four reciprocal pairs. Thus the residual graph has exactly three permutations, and Lemma 4.2 gives $P(175) = 3$.

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